

Spinning Wheels & Driving Motors!

Control DC Motors with Arduino Uno & L298N Motor Driver

1. Meet Your Robot's Core Components

The Brain: Arduino Uno

The microcontroller acts as the central brain. It processes logic, executes uploaded instructions, and directs signal commands to other components.

The Translator: L298N Motor Driver

Acts like a railway signalman! It acts as a power booster and directional bridge between the low-power Arduino and high-power motors.

The Muscle: DC Motor

The DC motor converts electrical energy into mechanical movement, spinning wheels to move your robot forward or backward.

CRITICAL SAFETY RULE: Direct Connection Risk!

NEVER connect a DC motor directly to Arduino digital output pins. DC motors draw significantly more current than Arduino pins can safely provide, which will cause permanent hardware damage to your microcontroller board.

2. Hardware Connections & Wiring Guide

To safely drive a motor, external battery power is routed through the L298N module while control signals come from the Arduino.

From Component	To Component	Purpose / Description
Arduino Pin 8	L298N IN1	Direction Signal Line 1
Arduino Pin 9	L298N IN2	Direction Signal Line 2
Arduino GND	L298N GND	Common Ground Reference (Mandatory for signals to work)
9V / 12V Battery (+)	L298N VCC (+12V)	Main Motor Power Source
Battery Ground (-)	L298N GND	Power Supply Ground Return
L298N OUT1 & OUT2	DC Motor Terminals	Power output to drive the motor windings

Think & Check: Why Common Ground (GND)?

Connecting Arduino GND to L298N GND ensures both devices share a common reference voltage. Without a shared ground connection, the directional control signals (HIGH/LOW) cannot be properly interpreted by the driver module!

3. Logic & Arduino Programming

The L298N uses two input pins (IN1 & IN2) to determine the direction of motor rotation:

Motor Logic Truth Table

IN1 (Pin 8)	IN2 (Pin 9)	Motor Action	Real-world Analogy
HIGH	LOW	Spins Forward (Clockwise)	Driving forward
LOW	HIGH	Spins Backward (Counter-Clockwise)	Reversing back
LOW	LOW	Stops Moving	Brakes applied / Idle
HIGH	HIGH	Stops Moving (Brake)	Brakes locked

Sample Code: Monsoon Wiper Pattern Challenge

```
// Arduino Motor Control Code
void setup() { pinMode(8, OUTPUT); // IN1 pinMode(9, OUTPUT); // IN2 }
void loop() { // 1. Spin Clockwise (Forward) for 2 Seconds
digitalWrite(8, HIGH); digitalWrite(9, LOW); delay(2000); // 2. Stop for 1 Second
digitalWrite(8, LOW); digitalWrite(9, LOW); delay(1000); // 3. Spin Counter-Clockwise (Backward) for 2 Seconds
digitalWrite(8, LOW); digitalWrite(9, HIGH); delay(2000); }
```

4. Troubleshooting Checklist

Motor Not Spinning?

- Check if the 9V external battery is connected to L298N VCC.
- Verify Arduino GND is firmly connected to L298N GND.
- Ensure power LED on L298N board lights up.

Motor Spinning Wrong Direction?

- Swap the OUT1 and OUT2 wire connections to the motor.
- Or flip `HIGH` and `LOW` values in your Arduino code.

Student Practice Sheet

Test Your Knowledge — Motor Control & L298N Integration

Name: _____

Date: _____

Q1. What is the primary role of the Arduino Uno in a robotic circuit?

Q2. Why is it unsafe to connect a DC motor directly to the output pins of an Arduino board?

Q3. Explain the function of the L298N Motor Driver module in your robot circuit.

Q4. What happens if you forget to connect the Arduino's GND pin to the L298N's GND terminal?

Q5. Fill in the states for standard motor operation:

- To rotate Forward: Pin 8 = _____ , Pin 9 = _____
- To rotate Backward: Pin 8 = _____ , Pin 9 = _____
- To Stop movement: Pin 8 = _____ , Pin 9 = _____

Q6. What command in Arduino C++ code is used to create time pauses between motor direction changes, and what unit of time does it measure?

Q7. If your motor turns backwards when your code sends a Forward signal, what are two ways you can fix this problem?